

GraphMap: Scalable Crowd-Sourced Global Vectorized HD Map Construction via Sparse Visual Graph Fusion

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Background

High-Definition (HD) Maps

- HD maps provide lane-level geometry and semantics (lane dividers, road boundaries, pedestrian crossings) beyond onboard perception.
- Conventional pipelines rely on dense LiDAR point clouds and heavy human annotation, so they are costly to scale and slow to update.
- Online HD mapping (e.g., MapTRv2 [1]) predicts local vectorized maps from onboard sensors in real time, but only near the ego vehicle.

Local mapping accuracy drops sharply with range (MapTRv2 mAP, %)

Vehicle · 30 × 30 m	82.7
Vehicle · 120 × 60 m	14.3
Infrastructure · 75 × 50 m	77.8
Infrastructure · 150 × 80 m	44.0

Main Challenges

Can we automatically build global vectorized HD maps from crowd-sourced, multi-agent data?

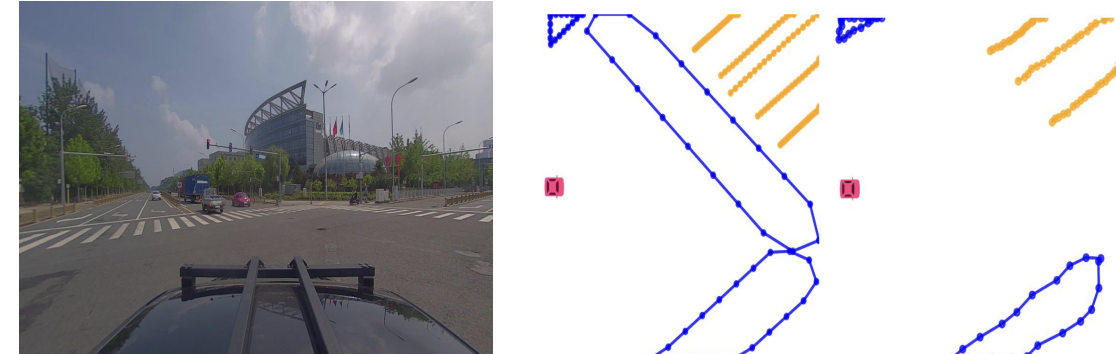
Why not using collaborative perception [2] methods?

- Ego-centric and fixed-range.** BEV tensors are bounded by the local field of view and a fixed perception range (e.g., 60 × 30 m).
- Infeasible at scale.** A 30 × 30 m region is a 100 × 100 BEV grid; a 3 × 3 km region becomes 10,000 × 10,000, too large to compute or transmit.
- Fragile under misalignment.** Global maps must fuse partially overlapping observations from different agents and times, with pose errors and time gaps.

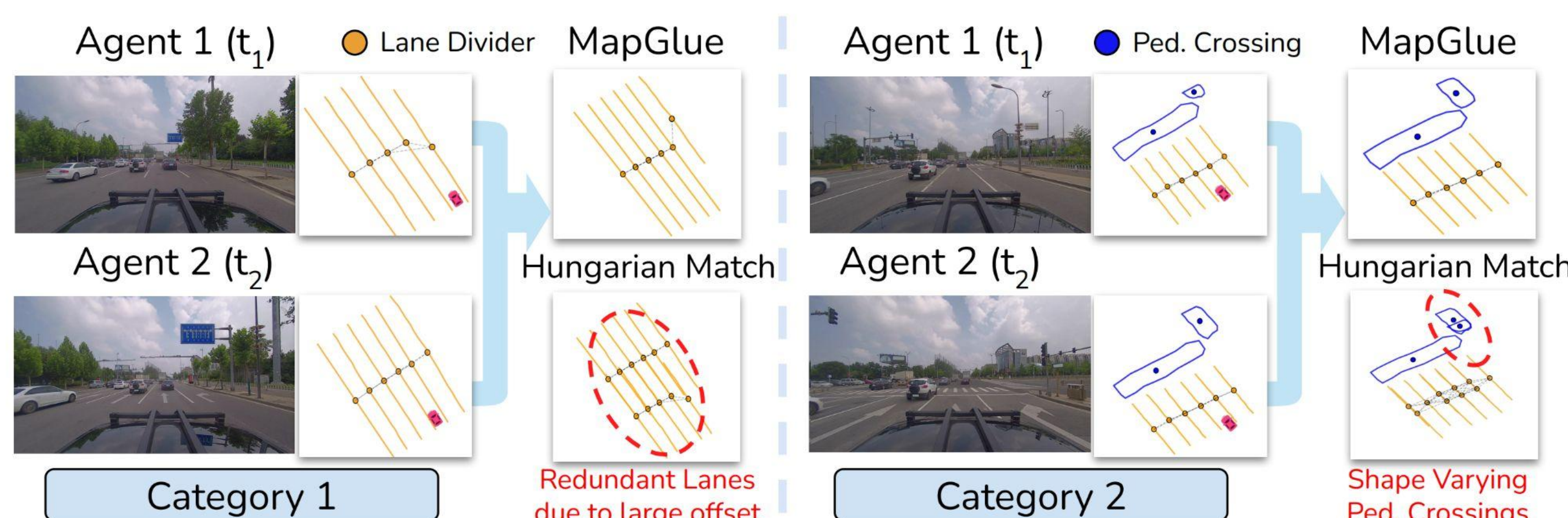
Limited Range



30m × 60m

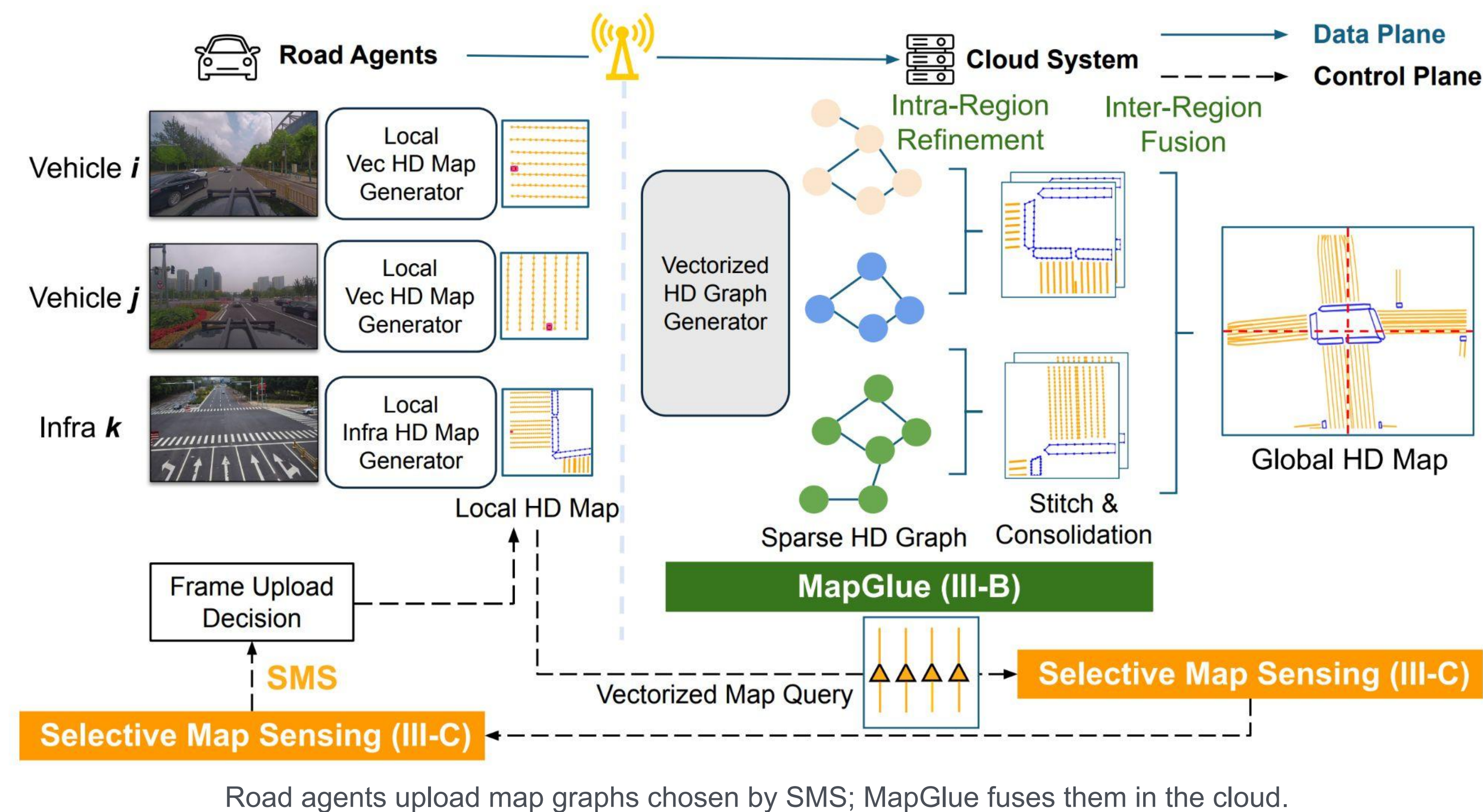


Two key observations in multi-agent map fusion



GraphMap: Crowd-Sourced Global HD Map Framework via Sparse Visual Graph Fusion

Goal: Scalable Crowd-sourced Global HD Map Construction Framework using Multi-agent Data



Sparse HD Graph Encoding

node $v = (x, y, \theta, \ell, \text{geometry}, \text{class})$

Each local HD map becomes a sparse semantic graph $G = (V, E)$. Nodes are map elements; edges encode geometric relations via Chamfer distance. Graphs are compact enough to upload over cellular or V2X links.

MapGlue: Type-Aware Graph Matching

$M[i, j] = M_{\text{node}}[i, j] \oplus M_{\text{edge}}[i, j], \infty \text{ across classes}$

Greedy approximation of the NP-hard maximum common subgraph problem. Category 1 may shift, under displacement-variance and heading constraints; category 2 must stay local, and matched polygons merge by union.

Selective Map Sensing (SMS)

$SV = \alpha \cdot \text{NewArea} + \beta \cdot \text{Consistency} + \gamma \cdot \text{Freshness} + \delta \cdot \text{DataType}$

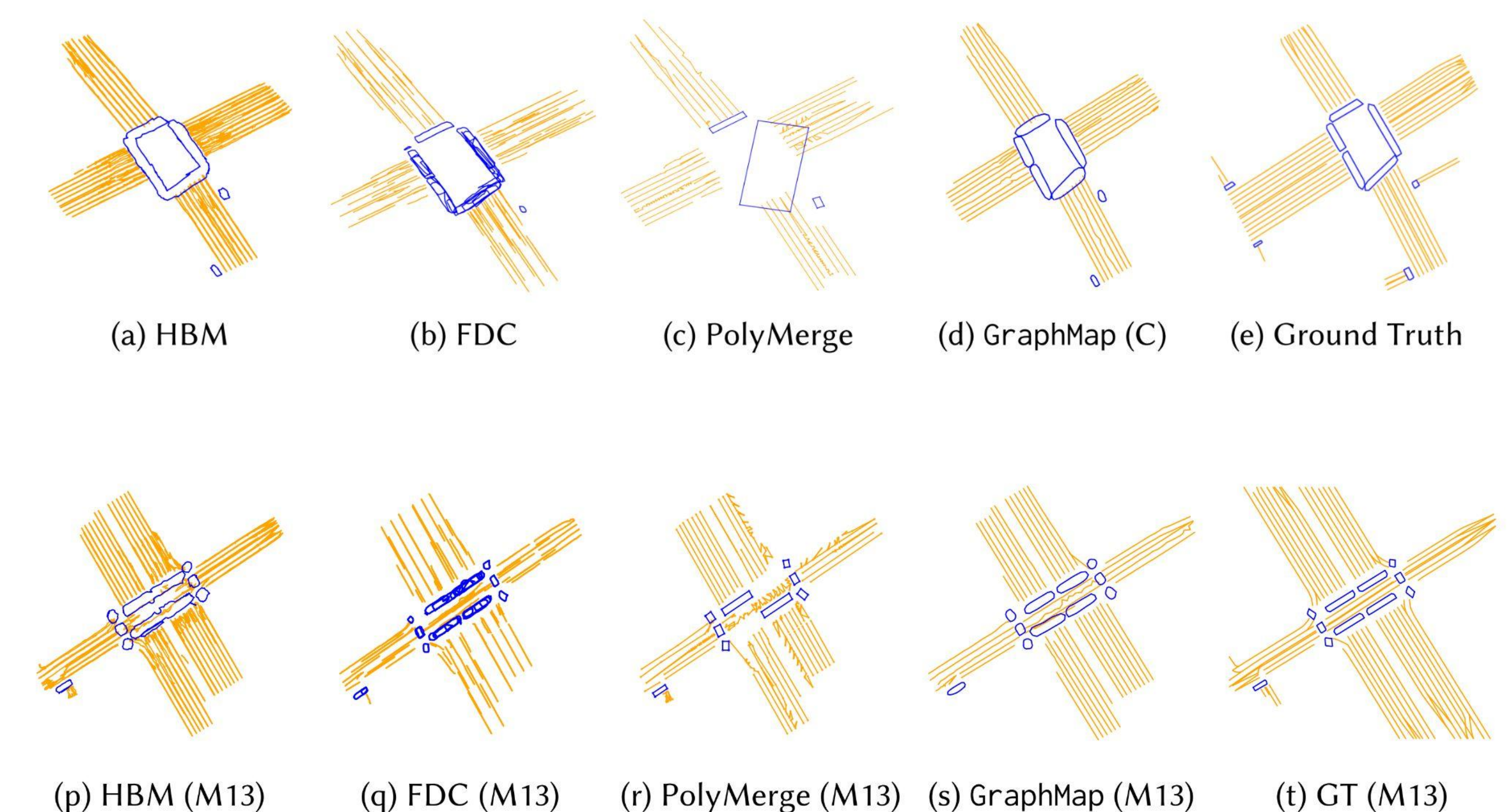
Double-ended: the cloud scores NewArea and Consistency against the global map; each agent scores Freshness and DataType locally. A feedback-delayed ϵ -greedy multi-armed bandit decides whether to upload each frame.

Evaluation Results

DAIR-V2X Maps: 06, 08, 09					DAIR-V2X Maps: 10, 13, 16				
Map	Method	AP _{ped}	AP _{div}	mAP	Map	Method	AP _{ped}	AP _{div}	mAP
Range: 300 × 300 m					Range: 300 × 300 m				
06	PolyMerge	0.148	0.085	0.117	10	PolyMerge	0.000	0.183	0.091
	FDC	0.202	0.027	0.114		FDC	0.310	0.003	0.157
	HBM	0.167	0.094	0.130		HBM	0.000	0.119	0.060
	GraphMap (C)	0.875	0.303	0.589		GraphMap (C)	0.857	0.416	0.636
	GraphMap (C+L)	0.875	0.320	0.598		GraphMap (C+L)	0.857	0.457	0.647
Range: 250 × 250 m					Range: 250 × 250 m				
08	PolyMerge	0.080	0.137	0.074	13	PolyMerge	0.618	0.081	0.399
	FDC	0.330	0.002	0.166		FDC	0.516	0.018	0.264
	HBM	0.000	0.229	0.114		HBM	0.575	0.063	0.319
	GraphMap (C)	1.000	0.502	0.751		GraphMap (C)	1.000	0.088	0.544
	GraphMap (C+L)	1.000	0.481	0.741		GraphMap (C+L)	1.000	0.163	0.581
Range: 300 × 300 m					Range: 250 × 250 m				
09	PolyMerge	0.769	0.058	0.413	16	PolyMerge	0.494	0.017	0.256
	FDC	0.594	0.046	0.321		FDC	0.450	0.012	0.231
	HBM	0.070	0.050	0.060		HBM	0.125	0.098	0.111
	GraphMap (C)	0.923	0.077	0.500		GraphMap (C)	0.833	0.099	0.466
	GraphMap (C+L)	0.923	0.166	0.545		GraphMap (C+L)	0.833	0.140	0.528

DAIR-V2X Map 06 (top) and Map 13 (bottom). **Orange**: lane dividers; **blue**: pedestrian crossings. GraphMap matches the ground truth; baselines duplicate lanes or distort crossings.

Visualization Results of Constructed Global HD Map



GraphMap achieves SOTA accuracy and builds global HD maps from multi-agent data at scale

References

- MapTRv2: An end-to-end framework for online vectorized HD map construction, IJCV'24
- V2x-vit: Vehicle-to-everything cooperative perception with vision transformer, ECCV'22
- PolyMerge: A novel technique aimed at dynamic hd map updates leveraging polylines, ICAR'23
- DAIR-V2X: A large-scale dataset for vehicle-infrastructure cooperative 3D object detection, CVPR'22
- V2XPnP: Vehicle-to-everything spatio-temporal fusion for multi-agent perception and prediction, ICCV'25